

1 Introduction

Remember to argue/justify anything you use in your answer. A good idea would be to include robust SE and a comparison of FE vs FD, for example

2 Outline

We've been handed a Cobb-Douglas function and its corresponding production function. We'll start by deriving an econometric model for the Cobb-Douglas production function using the provided panel data. We're allowed to pick any of the linear unobserved effects models for panel data from the course. We'll argue why we're using

Make the model and its regression

Estimator

Assumptions (efficiency/consistency)

Tests (when they're valid/invalid)

Results (betas værdier, f.eks., eller at u og x er korreleret. Hvad betyder dine resultater? Fortolk dem)

Important: Check for consistency and efficiency. Strict exogeneity is stricter than these.

Check for consistency:

$$\Delta y = \Delta x + \Delta u \quad \rightarrow \quad E(\underbrace{\Delta u_t}_{u_t - u_{t-1}} \overbrace{\Delta u_{t-1}}^{u_{t-1} - u_{t-2}}) = 0?$$

$$E(\Delta x \Delta u) = 0 \quad \rightarrow \quad \text{consistency}$$

3 Econometrics

3.1 Model

We've been given the following Cobb-Douglas production function:

$$y_{it} = \beta_K k_{it} + \beta_L \ell_{it} + v_{it}, \quad v_{it} = \ln A_{it}$$

It's also been mentioned that A contains both time-varying *and* time-invariant unobservable variables. With that in mind, we can unpack the model further and rewrite it to the following expression:

$$y_{it} = \beta_K k_{it} + \beta_L \ell_{it} + \underbrace{c_i}_{\text{Time-invariant}} + \overbrace{u_{it}}^{\text{Error term}}$$

3.2 Estimators

From here, we can estimate the model in a number of ways. One of which we'll demonstrate using Pooled OLS (POLS). For POLS to have consistency, it requires the following to be fulfilled, however:

$$E[k'_{it}c_i, \ell'_{it}c_i] + E[k'_{it}u_{it}, \ell'_{it}u_{it}] = 0$$

This assumption seems quite unlikely, however, because it relies on the notion that $E[k'_{it}c_i, \ell'_{it}c_i] = 0 \forall t$. This tells us that we must believe that there will *never* be a correlation between the variables (capital and labour) and the time-invariant across the 441 firms, regardless of time, and that's not something we'd expect. This is because something such as infrastructure or distance from work may affect labor and/or capital.

We're of the belief that at least for some t , there'll be a correlation such that $E[k'_{it}c_i, \ell'_{it}c_i] \neq 0$ holds. This assumption would make Fixed Effects (FE) and First Differences (FD) eligible candidates for estimating the econometric model instead.

3.3 Properties

The purpose of these methods is to perform a within transformation to eliminate the time-invariant, c_i , such that the estimator relies solely on the variation within each firm over time. In order to do this, we add a bar notation to denote time averages for our

model:

$$\bar{y}_i = \frac{1}{T} \sum_{t=1}^T y_{it}, \quad \bar{k}_i = \frac{1}{T} \sum_{t=1}^T k_{it}, \quad \bar{\ell}_i = \frac{1}{T} \sum_{t=1}^T \ell_{it}, \quad \underbrace{\bar{c}_i = c_i}_{\text{Unaffected by time}}, \quad t = 1, 2, \dots, T$$

Using our new notation, we can rewrite the model to include time averages in the following way:

$$\bar{y}_{it} = \beta_K \bar{k}_i + \beta_L \bar{\ell}_i + c_i + \bar{u}_i$$

We can then subtract the above from our econometric model in order to eliminate the time-invariant, and we've thus achieved the purpose of our within transformation:

$$y_{it} - \bar{y}_i = (k_{it} - \bar{k}_i)\beta_K + (\ell_{it} - \bar{\ell}_i)\beta_L + (\cancel{c_i} - \cancel{c_i}) + (u_{it} - \bar{u}_i) \quad (1)$$

For simplification, we can then denote the time-demeaning behaviour of our within transformed model using dots which yields the following:

$$\ddot{y}_{it} = y_{it} - \bar{y}_i, \quad \ddot{k}_{it} = k_{it} - \bar{k}_i, \quad \ddot{\ell}_{it} = \ell_{it} - \bar{\ell}_i, \quad \ddot{u}_{it} = u_{it} - \bar{u}_i$$

Within transformed model

$$\ddot{y}_{it} = \beta_K \ddot{k}_{it} + \beta_L \ddot{\ell}_{it} + \ddot{u}_{it}$$

We then have that the FE estimator of our within transformed model is exactly the same as the POLS but applied to the demeaned data.

$$\hat{\beta}_{FE} = \left(\sum_{i=1}^N \sum_{t=1}^T (\ddot{k}_{it} + \ddot{\ell}_{it})' \overbrace{(\ddot{k}_{it} + \ddot{\ell}_{it})}^{\text{1x1 Rowvectors}} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T (\ddot{k}_{it} + \ddot{\ell}_{it})' \overbrace{\ddot{y}_{it}}^{\text{1x1}} \right)$$

Instead of summing over i and t separately, we can stack $\ddot{k}_{it}$ s and $\ddot{y}_{it}$ s over first t and then i.

$$\begin{aligned} \hat{\beta}_{FE} &= \left(\sum_{i=1}^N (\ddot{K}_i + \ddot{L}_i)' (\ddot{K}_i + \ddot{L}_i) \right)^{-1} \left(\sum_{i=1}^N (\ddot{K}_i + \ddot{L}_i)' (\ddot{y}_i) \right) \\ &= ((\ddot{K} + \ddot{L})' \underbrace{(\ddot{K} + \ddot{L})}_{\text{NTx1 matrices}})^{-1} (\ddot{K} + \ddot{L})' \underbrace{\ddot{y}}_{\text{NTx1}} \end{aligned}$$

This shows that the fixed effects estimator is simply the OLS estimator on the transformed (demeaned) data.

4. Empirical analysis

4.0.1 (1) A description of the model selection process.

GOAL:

For the provided panel data setting we are choosing between different panel data methods, such as pooled ols (POLS), first differences (FD), fixed effects (FE) and random effects (RE) estimators. These estimators have different conditions for consistency and efficiency. In an ideal setting we want to choose an estimator that is as a first order priority consistent (1) and second order priority efficient (2).

INFORMATION GATHERING

To test these assumptions we run through a series of tests, Hausman, serial correlation and a strict exogeneity test. (AR1??), TODO describe the tests.

MOTIVATION

Problem: Our theoretical reasoning about the error term v_{it} representing TFP indicates that we should expect individual specific time invariant effects c_i . Dividing up the error term into $c_i + u_{it}$, which is typical for a panel data setting. F.ex location could be one effect that is time invariant for the individual company but has an impact on TFP and may be correlated with capital investment and labour employment.

Choice of estimator:

We observe that our test for strict exogeneity fails, meaning that our lead variable is significant and the regressors l_{it} and k_{it} are not strictly exogenous. This indicates that our standard fixed effects methods may be biased or inconsistent if used.

Normally we would address the presence of individual specific effects c_i by either removing them by demeaning with fixed effects FE, or first differences FD, or modelling them as random with random effects RE. Here we could use the Hausmann test to choose between demeaning FE, or partial demeaning RE, in our case the test shows us the FE may be

more suitable. However critically the validity of this test also depends on an assumption of strict exogeneity, which puts these results into question.

We should also be observant of the fact that while FD and RE depends on strict exogeneity, meaning that the error term u_{it} is uncorrelated with all past, present and future value of the explanatory variables k_{it} and l_{it} : $E[u_{it}|k_{i1}, k_{i2}...k_{iT}, l_{i1}, l_{i2}...l_{iT}, c_i] = 0 \forall t$, while POLS and RE could still be consistent under a weaker contemporaneous exogeneity CE, meaning that the error term u_{it} is uncorrelated only with the explanatory variables in the same time period t : $E[u_{it} | k_{it}, l_{it}, c_i] = 0$. The reason that contemporaneous exogeneity does not hold under FE or FD is that, If regressors are correlated with future error terms, the mean $(u_{it} - \bar{u}_{it})$ will also be correlated with regressors, making FE inconsistent. Similarly for FD we get the error $(u_{it} - u_{it-1})$ which includes the past error term u_{it-1} that may be correlated with k_{it-1} creating endogeneity bias. In other words the transformation of the error term removes the option of using CE.

The question now becomes, can we use contemporaneous exogeneity to make any of our estimators more consistent for POLS or RE?, and if so how do we weigh this against the benefit of demeaning individual specific effects away with RE or FD.

Discussing using RE under CE requires however additional assumptions namely that the individual specific effects must be uncorrelated with all explanatory variables $E[c_i | k_{it}, l_{it}] = E[c_i]$ (RE assumption) and no serial correlation in the error term u_{it} , $E[u_{it} | k_{it}, l_{it}, c_i] = 0$. Our AR(1) test for serial correlation and Hausman test, TODO: check assumptions AR1. However indicate that both these assumptions are violated. Since these test results again can also be called into question since we lack strict exogeneity, we may not just based on these completely rule out RE, however the fact that we empirically would expect some correlation between regressors and individual specific effects underlines the fact that RE may not be a good fit.

POLS may be even more unlikely to give consistent unbiased estimates, since it completely ignores c_i , it is generally highly unlikely that we don't have any individual specific effects for a panel dataset, and it goes against our empirical intuition of the problem. If we use POLS we assume that looking at the same company at two different time steps, generate just as independent data as if we looked at two completely different companies. The presence of c_i therefore makes POLS inconsistent and biased.

So how do we weigh the potential benefits with CE for the treatment of u_{it} against the downside of ignoring c_i that we can remove with FE or FD?

In a practical sense we are trying to determine if the omitted variables breaking the strict exogeneity condition are more likely to be firm specific and time invariant f.ex location and management, or firm specific and time variant f.ex how firms react to supply chain disruptions or changes over time. It seems empirically likely that firm specific traits are generally more constant than varying over time. This in combination with the results of our Hausmann test, although this may be unreliable, suggests that FE or FD while still biased may be a better option than POLS or RE.

So how do we choose between FE and FD?

Both FE and FD require strict exogeneity and will be biased. However between the two FE often makes more efficient use of the data since FD will remove one degree of freedom (a time step) for each individual, reducing the data available for estimation. If there is noise in the data or serial correlation, which our test, although unreliable, suggests there is, FD may inflate this problem. Between FD and FE, in our case, we opt for FE since we have $T=12$ years of data and it generally provides more efficient estimates, given that differencing can amplify measurement errors and serial correlation. Both FD and FE remove the time-invariant effect c_i . However, since strict exogeneity is violated, neither FD nor FE is fully consistent.

TODO discuss adding robust errors

- 4.0.2 (2)** A presentation and discussion of the relevant empirical results, such that the reader is able to understand and reproduce the steps carried out in the process as well as the conclusions made. The results must be presented in suitably formatted tables and/or figures. Irrelevant information should be left out of the tables/figures.
- 4.0.3 (3)** A discussion of whether the assumptions underlying consistent estimation and valid inference are likely to be satisfied/violated in the present empirical setting. This may include various forms of (mis)specification testing.
- 4.0.4 (4)** A clear conclusion to the main question and a discussion of the limitations of the approach used to reach the stated conclusion. Real-world examples of behavior/phenomena that may support or invalidate specific model assumptions should be related to the econometric model.
- 4.0.5 Concluding our hypothesis**

We set out to test the hypothesis that the Cobb Douglas production function exhibits constant returns to scale in this empirical setting. With our estimates for β_K and β_L derived from estimation on our regressors k_{it} and l_{it} . We will use a Wald test to test the hypothesis that we have constant returns to scale given the data and the Cobb Douglas production function for our inputs. This happens when the exponents sum to 1 meaning we are testing the following hypothesis: $H_0 : \beta_K + \beta_L = 1$ vs $H_A : \beta_K + \beta_L \neq 1$. A Wald test with a significance level of 5%, $\alpha = 0.05$ gives us the following result.

$$\text{With } \hat{\theta} = [\hat{\beta}_1, \hat{\beta}_2]^T, q = 1, R = [1, 1], V_{\text{robust}} = \begin{bmatrix} 0.0417^2 & 0 \\ 0 & 0.0300^2 \end{bmatrix} = \begin{bmatrix} 0.00174 & 0 \\ 0 & 0.0009 \end{bmatrix}$$

We have the following Wald statistic:

$$W = \frac{(R\hat{\theta} - q)^2}{RV_{\text{robust}}R'} = \frac{0.02286}{0.02286} = 8.662$$

We now want to compare to the χ^2 distribution with 1 degree of freedom for a 5% significance level:

$$\chi^2_{0.005,1} = 3.841$$

Since $8.662 > 3.841$ we reject our null hypothesis at 5% significance.

4.0.6 Limitations of the approach

- Breach of Strict Exogeneity assumption - A lead-variable test indicates that strict exogeneity fails, meaning none of our estimators including the chosen FE estimator is consistent, in addition the tests used to further make the choice of FE over other biased estimators are also reliant on an assumption of strict exogeneity and may also give unreliable results.
- Measurement Error - we are working with real world data and this is always a risk, we use FE over FD and this may help mitigate the impact of potential measurement error.
- Data selection - Our results are specific to the dataset of French manufacturing firms from 1968–79. Different industries or time periods could display different returns to scale for labour and capital.
- Variable missrepresentation - There is a risk that log deflated sales may not be an accurate measure of output or that log employment or log capital may not be accurate depictions of the inputs TODO: clarify var names and meaning.
- In a lecture not present in this year's course (lecture 5) a weaker exogeneity condition was discussed in relation to time lagged instrument variables, used in combination with FE and FD. This is outside the curriculum but might be an interesting option for further exploration.

4.0.7 Real world conditions

In the Cobb Douglas production function we have constant returns to scale when one additional unit of input for our input variables yields one additional unit of output. We observed omitted variable bias indicating that capital and labour are not in fact the only inputs required to produce an output, in many economic models we have additional factors f.ex human capital, land, oil etc. In that case CRS would imply that these exponents

together equals one, and therefore that capital and labour inputs on their own gives diminishing returns to scale DRS. Our result therefore makes theoretical sense as a more complex production function likely is to be expected in a real world scenario.

4.1 Empirical results

Table 1: **Table 1: Regression Results - Log of Deflated Sales**

Variable\Method	POLS	FE	FD	RE
Constant	1.54×10^{-8} (0.00497)	—	—	0.0000 (0.0162)
Log of Employment	0.6748 (0.0102)	0.6942 (0.0147)	0.5487 (0.0183)	0.7197 (0.0131)
t-value	66.4625	47.2447	29.9635	54.8243
Log of Capital Stock	0.3100 (0.0091)	0.1546 (0.0130)	0.0630 (0.0191)	0.1989 (0.0117)
t-value	33.9237	11.9311	3.3043	17.0239
R^2	0.914	0.477	0.165	0.642
σ^2	0.131	0.018	0.014	0.018
λ	—	—	—	0.887

Notes: Standard errors are in parentheses. The dependent variable is Log of Deflated Sales. R^2 is the coefficient of determination, and σ^2 represents variance. λ is reported only for the RE model.

4.2 Specification testing

Hausman Test			
Test Statistic	73.54	p-value	0.00
Exogeneity Tests			
Dependent Variable: Log of Deflated Sales	Beta	SE	t-value
Labour			
Log of employment	0.5681	0.0231	24.5458
Log of capital stock	0.1495	0.0134	11.1375
Union lead	0.1532	0.0225	6.8037
R^2	0.473	σ^2	0.016
Capital			
Log of employment	0.6479	0.0162	39.9359
Log of capital stock	0.0210	0.0231	0.9093
Union lead	0.1793	0.0258	6.9500
R^2	0.474	σ^2	0.016
Serial Correlation Test			
Dependent Variable: OLS Residual (e_{it})	Beta	SE	t-value
$e_{i,t-1}$	-0.1987	0.0148	-13.4493
R^2	0.039	σ^2	0.014

Table 2: Hausman Test, Exogeneity Tests, and Serial Correlation Test Results

5 Standard vs Robust variance estimation

The reason we'd prefer to use robust variance estimation instead of standard variance estimation for our fixed effects estimator is because it removes the need for FE.3 to be included for consistency. This in turn would alleviate us of some of the restrictions FE.3 otherwise would've imposed upon us such as homoskedasticity and no serial correlation. This makes it easier for us to make an estimation of our FE estimator since our data does in fact have serial correlation (see reference)

Category	Assumption
OLS Assumptions	
OLS.1	$E(X'u) = 0$ (exogeneity assumption)
OLS.2	$E(X'X)$ is full-rank (no perfect multicollinearity)
First-Differenced (FD) / Fixed Effects (FE) Assumptions	
FD.1 (= FE.1)	$E(u_{it} x_i, c_i) = 0, \quad t = 1, 2, \dots, T.$
FD.2	$\text{rank } E(\Delta X_i' \Delta X_i) = \text{rank} \left(\sum_{t=2}^T E(\Delta x_{it}' \Delta x_{it}) \right) = K.$
FD.3	$E(e_i e_i' x_i, c_i) = \sigma_e^2 I_{T-1}.$
Additional FE Assumptions	
FE.1	Strict Exogeneity: $E[u_{it} x_{i1}, \dots, x_{iT}, c_i] = 0.$
FE.2	Rank Condition: $E[\tilde{X}_i' \tilde{X}_i]$ has full rank $K.$
FE.3	Homoskedasticity (optional for efficiency): $E[u_{it}^2] = \sigma_u^2.$
Random Effects (RE) Assumptions	
RE.1	Strict Exogeneity: $E[u_{it} x_{i1}, \dots, x_{iT}, c_i] = 0.$
RE.2	Mean Independence: $E[c_i x_{i1}, \dots, x_{iT}] = E[c_i].$
RE.3	Homoskedasticity and No Serial Correlation: $E[u_{it}u_{is}] = 0$ for $t \neq s.$

Table 3: Summary of OLS, Fixed Effects, and Random Effects Assumptions